Software Requirements Specification for Software Engineering: subtitle describing software

Team 18, Team Magic
Allen Tong
Amos Edeaghe
Jacob Brodersen
Joshua Engel
Nehan Munir

October 20, 2026

Contents

1 Introduction **1**
- 1.1 Purpose . . . 1
- 1.2 Scope . . . 1
- 1.3 Stakeholders . . . 2
- 1.4 Regulations and guiding protocols . . . 3
- 1.5 Reasonable Inputs and Outputs . . . 4
- 1.6 Definitions and Acronyms . . . 5
- 1.7 Characteristics of Intended Reader . . . 6

2 System Overview **6**
- 2.1 Problem Statement . . . 7
- 2.2 Product Vision . . . 7
- 2.3 System Context . . . 7
- 2.4 User Characteristics . . . 8
- 2.5 System Constraints . . . 9

3 Stakeholder Requirements **9**

4 User Personas **9**
- 4.1 Persona 1 . . . 9
- 4.2 Persona 2 . . . 10
- 4.3 Persona 3 . . . 11

5 Use Cases **12**
- 5.1 UC1: Round Setup . . . 12
- 5.2 UC2: Tasks and Answer Checking . . . 13
- 5.3 UC3: Scoring and controls . . . 15
- 5.4 UC4: Saving and Recovery . . . 16
- 5.5 UC5: 3D Visualization . . . 18
- 5.6 UC6: Guided Learning and Tutorial . . . 20
- 5.7 UC7: Create Account . . . 22
- 5.8 UC8: Daily Challenge . . . 23

6 Functional Requirements **25**
- 6.1 Formal Model . . . 25
  - 6.1.1 Game Configuration . . . 25
  - 6.1.2 Round State . . . 26
  - 6.1.3 Task Evaluation . . . 27
  - 6.1.4 Timed Scoring . . . 28
  - 6.1.5 Formal Properties . . . 28
- 6.2 Requirements . . . 29

7 Data Requirements **34**
7.1 Major Data Entities . . . 34
7.2 Data Dictionary . . . 35
7.3 Privacy and Data Retention . . . 36

8 Non-Functional Requirements **37**

9 AI and Machine Learning Requirements **41**

10 Requirements Traceability Matrices **41**

11 Rationale **43**
11.1 Scope . . . 43
11.2 Functional Requirements . . . 44
11.3 Data Requirements . . . 45
11.4 Nonfunctional Requirements . . . 46

12 Likely Changes **47**

13 Unlikely Changes **48**

14 Appendices **49**
14.1 Appendix A: Additional Materials . . . 49
14.2 Appendix B: Development Plan . . . 49
14.2.1 Phase 1: Proof of concept . . . 49
14.2.2 Phase 2: Core game loop implementation . . . 49
14.2.3 Phase 3: Improved 3D visualization and Navigation . . . 50
14.2.4 Phase 4: Learning phase . . . 50
14.2.5 Phase 5: Online feature phase . . . 51
14.2.6 Phase 6: Recovery phase . . . 51
14.2.7 Phase 7: Stretch goal implementations . . . 52
14.3 Appendix C: Values of Auxiliary Constants . . . 52
14.4 Appendix D: Generative AI Disclosure . . . 52
14.5 Appendix E: Reflection . . . 53

Revision History

Date	Developer(s)	Notes
October 19, 2026	Nehan Munir	Added SRs, Personas, UCs, and (U)LCs and updated sections 11, 12, and 13
October 19, 2026	Allen Tong	Added Sections 6, 7, 9 and updated sections 11, 12, 13
October 19, 2026	Amos Edeaghe	Added Sections 1, 10 and updated sections 11, 12, 13
October 19, 2026	Joshua Engel	Added Sections 2, 8 and updated sections 11, 12, 13
October 19, 2026	All Team Members	Added Reflection

1 Introduction

This document specifies Magic-8, a free-to-play, web-based 3D puzzle game built around an 8×8×8 magic cube. It states the purpose, users, and project goals: to provide an engaging, navigable magic cube experience in which players complete tasks to practice addition and subtraction in base 8, supported by an introductory tutorial, guided hints, online competitive features, and task-based problem solving.

The work is based on the provided IEEE SRS template provided in the course **?** with reference to the ISO/IEC/IEEE 29148:2018 template **?**, which justifies the SRS structure, and ISO/IEC/IEEE 15288:2023 **?**, which provides the broader life-cycle context underlying modern requirements engineering practice. Additional sections were added in the form of regulations and guiing protocols, and reasonable inputs & outputs to increase the team's refinement of the project scope and definition.

1.1 Purpose

The project is intended to create a game involving an 8x8x8 magic cube that can be navigated by players through task completion to practice addition and subtraction in base 8. The intent is to solve the lack of engaging resources available for learning and practicing mathematics in non-decimal number bases as alternative bases. Such as base 8, are rarely taught before post-secondary education.

The following sections will outline the scope, functional requirements and the non-functional requirements that the team will adhere to produce a game that is both educative and fun.

1.2 Scope

In-Scope and functionality

The project known as Magic-8 will be a free-to-play web game where users will be presented with a 3D visualization of an 8×8×8 magic cube separated into slices. Users receive a tutorial on addition and subtraction in base eight, a starting cell, and at least 8 tasks that, by solving the problems within, will navigate them through the cube as they move through its layers. Users will gain points for correctly solving the problems in a task; these points will be tied to a user's personal account.

Intended users

Magic-8 primarily appeals to recreational math enjoyers who would enjoy a challenge while practicing a different base, and high school students and their teachers who gain a way to engagingly experience an unfamiliar base. The goal is to create a system that can not only be used as a fun way to improve one's skill in addition and subtraction in bases but would also act as a ready-made introductory classroom tool for engaging students, an experience that is helpful for aspiring STEM students due to how important base 8 and working with new number bases is in systems such as Unix, the C programming language, etc., with Magic-8 serving as an early introduction. As such, the improved interest in

numeracy stemming from a broader appreciation for alternative number bases benefits all those who would appreciate the greater interest, such as post-secondary STEM professors, educational technology platforms, and society as a whole.

Out of Scope

The tool is aimed at being fun and educational; it is not intended to replace a curriculum or full octal tutorial; it is strictly web-based with no plans for mobile or offline development. It supports octal only (no other bases, no 2D squares, no 4D tesseracts, no alternative rule-sets, no automatic cube generation, no player puzzle generation, no decimal mode, although these may be considered as stretch goals after project completion). Cubes of order other than 8 and 16 are excluded because they do not satisfy all 13 lines.

Explicit assumptions

The project assumes the following as true for the success of the project:

- A valid order 8 magic cube is available.
- New valid cubes can be created be it algorithmically or through rotation of initial cube.
- A user will not be able to memorize the positions of the 512 cells that make up the cube to bypass task interaction.
- Users have browser capable devices and internet access.
- Users posses basic numeracy.
- The chosen implementation software's open source licenses are compatible with the distribution method.
- Tasks are all solveable and their problems can be accurately conveyed to intended users.
- Users are willing to engage with an octal system.

1.3 Stakeholders

A stakeholder for the project is defined as any independent party with an interest in the sucess of the game or is affected by it. The team will categorize stakeholders following the SRS IEEE template however due to our team acting as the development team the development team category will not apply.

- **Customer:** The game is free to play; however, technological educational websites who may choose to host the website to increase their catalog. The aim is also for high school teachers to adopt the game as a way of maintaining students' attention in fun and educational activities that are still classroom appropriate.

- **End Users:** The end users include the recreational math enthusiasts who play the game for intellectual stimulation and the novelty of magic cube mathematics; they may already be familiar with octal, but their main goal is entertainment over education. Another primary end user are the high school students who play the game to practice basic octal in a way that they find engaging.

- **Project Sponsor:** The capstone supervisor endorses and provides the context and the initial goals which the project team should meet; their involvement means the project needs to meet its stated goals within the capstone timeline and demonstrate its recreational value as can be seen from their history in proposing recreational math games. The supervisor also provides institutional insight that guides how the project is specified and implemented, with suggestions on the ways cubes can be generated, presented to the user, and how difficulty and tasks should be addressed.

- **Administrators:** The administrators for this project would be in charge of managing the online components of Magic-8 such as the user account and login information and are concerned with ensuring the privacy and integrity of the user data. In a similar vein, the regulatory and ethics bodies that set rules for data minimization and privacy protection policies are concerned with the game meeting all their related policies.

- **Society:** Society in this context refers to the general public; one of the game's goals is to educate and breed interest in society for mathematics. This is intended to increase the numeracy of the general public and increase interest in programs such as mathematics and engineering. As it is intended to reach the general public, there is also interest from the families of users in the game being free, safe, private, and inclusive to those from any background and conscious of the health issues that might be faced by the users, requiring careful thought into how the visuals, inputs, and outputs are handled to make them non-safety critical or triggering in any way. The product will likely be used by more vulnerable members of society such as teenagers, and thus care must be given to ensuring their information and safety are not compromised from their gaming experience.

1.4 Regulations and guiding protocols

The project must satisfy all the following rules and regulations:

Standard / Protocol	Description
ISO/IEC/IEEE 29148:2018	International standard defining the structure and content of requirements specifications, used to justify the organization of this SRS.
ISO/IEC/IEEE 15288:2023	International standard describing software and systems life-cycle processes, providing the broader engineering context for requirements practice.

Standard / Protocol	Description
WCAG 2.1 Level AA	Web Content Accessibility Guidelines defining accessibility requirements for web content; the game targets Level AA for keyboard navigation, contrast, and text resizing.
W3C WebAssembly	World Wide Web Consortium standard for portable binary code executed in browsers; the game is packaged via pygbag to run as a WebAssembly module.
W3C HTML5	World Wide Web Consortium standard for web page structure and browser interaction; the game interface must conform to it.
GDPR	General Data Protection Regulation governing the processing and protection of personal data for users in the European Union.
PIPEDA	Personal Information Protection and Electronic Documents Act governing personal data handling for users in Canada.
COPPA	Children's Online Privacy Protection Act governing data collection from users under 13; applies if the game is made available to children below that age.
FERPA	Family Educational Rights and Privacy Act governing student education records; applies if the game is integrated into formal classroom settings.
Open-source licenses	Licensing terms of libraries such as pygame-ce, pygbag, pylint, pytest, and mypy; the game must comply with all dependency licenses.

Table 1: Regulations and Guiding Protocols

1.5 Reasonable Inputs and Outputs

The following outlines some of the projects expected user inputs:

Input	Description
3D navigation commands	Rotation, zoom, and movement through cube layers via keyboard or mouse.
Voxel selection	Selecting a specific voxel number within the cube.
Answer submission	Submitting a selected voxel or value as the answer to the current task.
Tutorial interactions	Advancing through, replaying, or skipping the tutorial.
Account credentials	Username and password for account creation and login.
Round start	Player begins a new round, triggering task generation.

Input	Description
Database state	Stored accounts and statistics, retrieved from local or online databases.
Authentication responses	Responses from third-party identity providers, if integrated.
Asset and configuration files	Game assets, cube visuals, and configuration data loaded at startup.

Table 2: Project Inputs

The following outlines some of the projects expected user inputs:

Output	Description
3D cube rendering	Visual display of the 8×8×8 magic cube with visible and hidden voxels.
Slice and voxel highlighting	Visual differentiation of slices, selected voxels, and active lines.
Task progress display	Current task, progress within the round, and remaining tasks.
Correct/incorrect indication	Clear feedback on whether an answer was correct or incorrect.
Error explanations	Explanation of mistakes users make while playing.
Score and statistics display	Points earned, times, accuracy, and other gameplay statistics.
Tutorial content	Interactive instructional content for learning to play and for octal basics.
Account confirmations	Confirmation of account creation and login.
Local database updates	Stored accounts, statistics, and settings persisted on the user's device.
Online database updates	Synchronized accounts and game statistics on the server.

Table 3: Project Outputs

1.6 Definitions and Acronyms

This section provides the meaning of the different words and phrases used in the domain of the problem.

Term	Definition

Base 8	A number system using the digits 0 to 7 where the value for each digit in decimal can be found by multiplying by 8 to the power of n where n represents the number of positions from the right.
Octal	Another name for base 8.
Magic square	A square array of positive integers where the sum of each individual row, column, or main diagonal is the same.
Magic cube	The 3-dimensional equivalent of a magic square where the sums of each row, column, pillar, main diagonal, or triagonal is the same.
Magic constant	The fixed sum obtained when doing sums in any row, column, or main diagonal in a magic square.
Most perfect magic cube	A magic cube where all rows, columns, pillars, diagonals, triagonals, and broken space diagonals add up to the magic constant.
Triagonal	A line or set of cells connecting two opposite vertices through the interior of a cube and hence moves across all three axes.
Broken space diagonal	A diagonal that wraps around the cube's boundaries.
Voxel	A single 3D cell within the magic cube.
Slice	A 2D display providing all the voxels within a certain cross section of the magic cube.
Round	A single playthrough of the game consisting of 8 sequential tasks.
Starter	The initial randomly chosen visible voxel that acts as a starting point for the player to start solving tasks.

Table 4: Definitions and Acronyms

1.7 Characteristics of Intended Reader

The intended readers of this document are expected to be able to read, review, maintain, or implement the SRS. As such, they are expected to have knowledge equivalent to the final year of a software engineering student or a student of an equivalent degree. This includes an intermediate level of experience with a programming language such as Python, a beginner level understanding of 3D graphics, game development, WebAssembly, and CI/CD.

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

2.1 Problem Statement

Magic cubes provide an opportunity to explore numerical patterns and mathematical relationships in 3-dimensions. However, understanding these structures can become challenging due to limited sources as well as having to use static 2D images to understand a 3D pattern.

The goal of this project is to create a recreational mathematics game that allows those interested in these magic cubes as well as non standard number bases such as base 8 to further their understanding. This is planned to be a web-based game in which users will be able to complete certain challenges on a 3d representation of an 8x8x8 magic cube using base 8 mathematics to find patterns associated with a magic cube. This will include adding values to certain sections of the cube to complete patterns of magic cubes moving through 8 stages of difficulty while continuing to learn about these mathematical structures.

2.2 Product Vision

The vision for Magic-8 is to provide an accessible and engaging recreational mathematics application to allow interested users to further their understanding of non base 10 number systems as well as mathematical patterns found in objects such as magic cubes. The system will allow for simple interaction with the magic cube to allow users to see its many different views as well as help educate them in the patterns they will be using throughout the challenges.

This will be able to operate on supported desktop and laptop web browsers and will provide some account-based storage to track the users performance and rank them on a leaderboard. The system is planned to focus on maintainable and modular system design to potentially allow for different number bases in the future as well as ease of use for the users of the application.

2.3 System Context

- User Responsibilities:
  - Provide the information required for supported account operations.
  - Select puzzles or activities and provide numerical inputs through the interface.
  - Use the available controls to manipulate the cube, select cells, and edit cell values.
  - Review validation messages and feedback provided by the application.
- Software Engineering Responsibilities:
  - Validate user inputs and identify invalid or unsupported values.
  - Process supported cube interactions and maintain the current puzzle state.
  - Present the cube and associated numerical information in the user interface.
  - Provide feedback about input validation and the outcomes of supported operations.

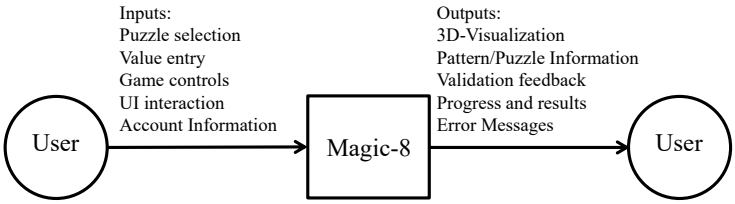


Figure 1: System Context

  - Communicate with the online service to authenticate users and save or retrieve designated user data.
  - Inform users when account or storage operations fail or cannot be completed.

- Web Browser Responsibilities:
  - Execute the delivered application and provide the environment in which its interface is displayed.
  - Render the two-dimensional interface and three-dimensional cube visualization.
  - Relay user interactions to the application.

Magic-8 is intended for recreational and educational exploration. This means it will not be operating in a highly safety or mission-critical enviornment. The application provides a simple way to interact with slighlty more obscure mathematical ideas rather than any critical operations.

2.4 User Characteristics

The intended users are individuals who are interested in recreational mathematics, numerical puzzles and structures, and non standard number bases. Users may range from beginners who may have just started learning base 8 and never encountered magic cubes or vice versa, to experienced mathematical enthusiasts interested in trying a new challenge.

Users are expected to have basic computer skills, including the ability to navigate a web browser, use pointer-based controls, and enter numerical values. No programming knowledge

is required for use. Only basic mathematic understanding of addition and subtraction is expected as the application can help to teach the subjects it covers.

2.5 System Constraints

Magic-8 shall be designed to run within desktop and laptop web browsers without requiring an installation. The functionality will initially be developed in python as well as supported python libraries such as pygame. It will then be translated into webassembly code for execution in browser. This means that all python code must be translatable with the chosen tool into webassembly including anything used for the user interface of the application.

The system shall also support online account functionality, requiring network communication with the selected service for remote data operations. The specific service, supported web browsers, and rendering technologies are yet to be established. This will happen as implementation decisions are finalized.

3 Stakeholder Requirements

SR1 This program shall validate the stakeholder's answers in terms of both position and octal answer.

SR2 This program shall visualize the current octal magic cube being solved by the stakeholder.

SR3 This program shall teach stakeholders how octal number works.

SR4 This program shall teach stakeholders how magic squares and then cubes work.

SR5 This program shall verify if the stakeholders satisfied all the requirements of a level.

SR6 This program shall show stakeholders different cubes with tasks of the same difficulty at the same level.

SR7 This program shall time stakeholders and show them their current and final score.

SR8 This program shall allow stakeholders to access it deployed on a free web-hosted website.

4 User Personas

4.1 Persona 1

- Name: Scotty Thompson
- Age: 16

- Occupation: Student
- Location: Toronto, Canada
- Description: Scotty is a Grade 10 student who stumbled upon this program when he was learning octal numbers in his Computer Technology class. His teacher took a glance at the program and recommended using it to study for the octal/hexadecimal test. The only understanding he has of the octal system is from his computer technology course, which he pays attentions to. but is lost in his train of thoughts when learning about concepts that don't apply to everyday life.
- Goals:
  - With a basic undertanding of octal numbers, Scotty wishes to learn how to do them faster (like a multiplication table), so he doesn't get stuck on test questions requiring addition/subtraction.
  - He doesn't want a simple math checker, he wants more information on why he got the answer wrong, alongside a visual demonstration on how to get the correct answer.
- Frustrations:
  - He doesn't care about the "magic" and intrinsicality of the magic cube, as he feels like he has no use of learning about these patterns.
  - He will get bogged down if shown literature about magic cubes as he is using this program for only one goal.
- Technical Ability: He is a gamer who has played browser games since he was a child and is able to quickly click and type.

4.2 Persona 2

- Name: John Fye
- Age: 73
- Occupation: Retired (Former Stenographer)
- Location: Hamilton, Canada
- Description: John is a former court stenographer who recently got a kick out of learning about John Hendricks's work. He wishes to test out his ability to see the patterns in magic cubes as he has been doing base-10 magic square puzzles in his puzzle book for the last 20 years.
- Goals:

  - He wants to learn more about octal numbers, as he has a very basic understanding about it, but a lot of Mr. Hendricks's work finds patterns in different bases, and he feels like learning about octal will help him appreciate Mr. Hendricks's accomplishements, such as order 16 perfect magic tesseract.
  - He wants to discover the patterns in the cube, such as how you can use a starting number to find other numbers (such as the complimentary).
  - He really wants to dive deep into

- Frustrations:
  - The only thing he currently knows about octal is how it cycles from 0 to 7, he never did any math with it.
  - He struggles to look at a screen for a long time and is usually zoomed in. He takes extensive breaks, which might lead him to forgetting where he left off and what was his train of thoughts at that point.

- Technical Ability: He was a court stenographer, which allows him to type fast. However, he never really used a computer before, which will be a problem when he is using a mouse. He also doesn't want to look at a mostly white screen when using the program.

4.3 Persona 3

- Name: Charles Dean
- Age: 23
- Occupation: Masters in English
- Location: Halifax, Canada
- Description: Charles is someone that starts off their day playing the "Dles", which is a collection of single-level browser games that refreshes daily, like Wordle. He is tired of the flood of general knowledge trivia games, and wants to play a game that tests his brain without needing to know stuff beforehand.
- Goals:
  - He wants a daily game that he can complete as part of his daily rotation of games (shouldn't be too long).
  - He wants the game to run on his browser with minimal load times, not really interested in getting stuck due to the 3D visualization.

- Frustrations:

  - He isn't really interested in math, so might lose interest if the puzzles are too complex.
  - He needs a hook (either visually or with flair) that ensures that he plays it more than once as he doesn't start off interested in the subject matter.

- Technical Ability: He is very familiar with browser games and while he hasn't done math since high school, he does remember an inkling of what hexadecimal is from his high school computer course.

5 Use Cases

5.1 UC1: Round Setup

Primary Actor

User

Preconditions

- Browser game is deployed on an accessible website
- Game has access to database containing cube configurations
- No rounds is currently in progress

Main Success Scenario

1. User presses START
2. System wipes screen clean and pings cube database wih appropiate query
3. Cube database sends the values of the cells alongside with a cube number (as an identifier)
4. System creates the 3D visualization and discretely populates visualization cells with its correct, hidden values
5. System prompts user to choose the scoring system (decimal or octal)
6. User clicks on the scoring system they want and its confirmation
7. System selects the appropiate scoring system
8. System randomly populates a single visualization cell with its correct octal and has a final popup describing the next level

Alternative Flows

- If Cube database doesn't ping back, the System will already have one default cube formation stored locally in its code, so that cube formation will be used instead. The System will continuing pinging the database every 30 seconds to see if the database is active and notifiy its developers the first time it doesn't get an answer.
- If the 3D visualization fails, either fall back to a simpler visualization method or show an error.
- If the scoring system fails, continue with the steps after displaying a message about how scores isn't going to be tracked for this round.

Postconditions

- All errors and log-type messages will appear as notifications starting from the bottom on a corner.
- There is a black 3D visualization of a cube in the centre.
- There is a notification showing '512 of 512 cells loaded' and there is a single octal number on the cube.
- On one of the upper corners, there is a scoring component consisting of an unstarted timer and a score integer.

5.2 UC2: Tasks and Answer Checking

Primary Actor

User

Preconditions

- The system should be set up according to the postconditions of UC1

Main Success Scenario

1. User chooses to begin the levels by clicking START.
2. System displays a textbox asking the user to select the complement of the single populated cell
3. User moves the 3D visualization using the chosen movement method in order to locate the complement
4. User clicks on the cell representing the complement, thereby selecting it
5. System flashes white and displays the properties of the intial populated cell and a textbox for the answer

6. User enters in the complement's value in the Addition/Subtraction Box
7. System alerts the scoring syetem by sending a query with the level of the task that was just finished
8. System shows the user a level overview with a difficulty slider
9. User chooses difficulty
10. System populates the cells according to the difficulty selected and displas a START button
11. User clicks START and selects the cells that corresponds to the half-orthogonal for the starting cell
12. User clicks the DONE button
13. System displays the Addition/Subtraction Box for selected cells
14. User enters in the octal values and clcks DONE
15. System queries the scoring system with information of the level just completed
16. Previous steps repeat for the remaining steps: orthogonal, diagonal on continuous plane, diagonal not on continuous plane, another diagonal not on continuous plane, triagonal, split triagonal
17. System asks scoring sytem for final score
18. Scoring sstem calculates final score
19. System ends after displaying a final score and buttons to navigate to other sections

Alternative Flows

- User is able to pull up level overview when they are in the level by selecting an arrow poking out of the side if they are stuck
- If the User clicks DONE after selecting the wrong cells, the game will flash red and deselect the selected squares, and siplays a message saying they selected the wrongs squares
- When the user inputs the wrong answer in the Addition/Subtracton box, the box will flash red and clear the incorrect answer, letting them try again. The System will also notify the scoring system that an incorrect answer has been given.
- If the 3D visualization is stalling (e. due to the user running out of memory), the System shows a toggle to switch to the 2D view of the cube

- If the user is signed in, the are able to save the score if they want to

Postconditions

- System should display the final score front and centre
- There should be an option to either create a new account (while being able to save the current score), sign in, or view account information
- There are options displayed to either start a new run, go to the tutorial, or enter other modes

5.3 UC3: Scoring and controls

Primary Actor

System

Preconditions

- The system is currently active and in a viable state of execution
- The scoring system is an instance which is volatile and doesn't keep track of information
- The scoring system is well integrated with near-instantaneous signals from the system
- The scoring system is able to use other APIs to ensure that its time signals is correct

Main Success Scenario

1. System queries the scoring system with level number
2. Scoring system initializes a stopwatch, error counter and correct scoring function
3. System pings the scoring system, telling them that the level has started
4. Scoring system starts the timer and listens for signals indicating errors or completion
5. System pings for an error
6. Scoring system increments error counter
7. System pings for the completion of the level
8. Scoring system runs appropiate scoring function with time and error counter as inputs
9. Scoring system returns the score and its instance terminates

Alternative Flows

- If the user selected a different scoring base or level, the system (not scoring system) will add the corresponding values for each addition:

Level	Decimal	Octal
1	10	14
2	20	30
3 and 4	30	44
5 and 6	40	60
7 and 8	60	110

- If the user is able to enter in additional numbers, they are rewarded 10 points for desimal and 14 points for octal.
- If the user clicks STOP/EXIT, the scoring system instance is terminated without any return
- If the user clicks PAUSE, the stopwatch in the scoring system will pause and keep track of the time paused (so it doesn't mess up when checked with eternal APIs), but the scoring system instance will continue running

Postconditions

- The score given can be verified as base score * (1350/seconds) + score for correct cells/octals for an individual level
- The system's internal final score is incrememented between levels
- At the end, the system shall show the internal final score (verified as last internal score + level 8 score)

5.4 UC4: Saving and Recovery

Primary Actor

User

Preconditions

- System should be in a valid state of execution
- User database should be active (in the sense that it exists and connectable) and triggerable
- The connection between the database and system should be closed.

Main Success Scenario

1. User selects cells and pressed DONE
2. System saves which cells were selected
3. System verifies that the correct cells were selected
4. System saves that the correct cells were selected
5. System calculated how much points the user gets for the selection
6. System saves the number of points
7. System displays the Addition/Subtraction box based on saved game state
8. User completes level
9. System caculates and saves final score
10. User chooses to save the score
11. System pings user database with user information
12. User database pings back (saying that it is active)
13. System shares final score with user database
14. User database saves score
15. System closes the connection with the user database
16. User clicks to return to main menu
17. System refreshes its information via user database
18. System displays the main menu with user information (including latest/highest final score)

Alternative Flows

- If a save fails (either internal or with a database), the system will retry 3 times. If the system still fails to save, it will notify the user and warn that any further progress might not be saved. The user is free to keep playing and the system will retry saving either every 30 seconds or when the user reaches the net save point (whichever comes first).
- If the program crashes, after exhausting all possible recovery options, the system will save the last recoverable state in the browser cache. It will then recommend the user refresh or close the browser page. The next time the user opens the page, it will ask the user if it wants to load the previous progress or start anew.

Postconditions

- The connection between the user database and system should be closed
- The schema and integrity of the user database should be equivalent to its precondition
- The system shouldn't have any saved data about internal levels (such as level scores and cell selection) after finishing all the levels
- The browser cache would optimall always be empty unless there has been a crash and the user hasn't returned to the game

5.5 UC5: 3D Visualization

Primary Actor

System

Preconditions

- System is in a valid state of execution
- 3D visualization is a plugin (removing it just causes the system to use another visualization)
- 3D visualization doesn't have any data inputted into it (it starts anew everytime a cube is selected)

Main Success Scenario

1. System gets the cube values and number from the cube database
2. 3D visualization starts to initialize in the middle of the screen
3. System chooses a starting position
4. 3D visualization filled with the correct octal number at that position
5. System repositions the cube to the centre of the screen
6. 3D visualization centers the model at the filled cell
7. User moves mouse across a face of the cube
8. 3D visualization rotates accordingly across its face due to mouse
9. User types in a coordinate in a navigator bar
10. System repositions the cube to the centre of the screen

11. 3D visualization centers the model at that coordinate
12. User presses down on the "s" or "down" key on the keyboard
13. 3D visualization rotates down accordingly across its face due to keyboard
14. User drags mouse on an area off the cube
15. System repositions towards the mouse (so that the cube is off centre)
16. User scrolls down on the mousewheel
17. 3D visualization moves along the z-axis, so that the second and next layers of the cube is exposed
18. User selects a cell
19. System saves which cell has been selected and checks its correctness
20. System tells the 3D visualization which cell has been selected and what colour it should be
21. 3D visualization keeps the cell and previous cell as a bright white light source
22. User enters in number in Addition/Subtraction box
23. System checks correctness of the octals inputted
24. System tells the 3D visualization what to project onto the cell's surface
25. 3D visualization shows the correct octals the user inputted
26. System calculates level score
27. 3D visualization keeps filled cells

Alternative Flows

- If the user types in a coordinate that would belong on another layer, the system will reposition the cube to the centre of the screen and the 3D visualization centres on the selected xy coordinates. It will then move through each layer until it arrives at the correct z value.
- If the user is selecting the wrong cell (ex. as part of a triagonal), the system will inform the 3D visualization that the cell chosen is wrong. The 3D visualization will make the cell chosen red.
- If the user is on a non-face and tries to rotate the cube, the user will be able to see the missing layers either behind or on the R/LHS of the camera, but they will be transparent so the person is aware of where they are located in the cube.

- If the system is calculating final scores, the 3D visualization isn't needed, so that instance of the 3D visualization plugin is terminated.

Postconditions

- The 3D plugin shouldn't have any data associated with it (such as information about which cells are filled)

5.6 UC6: Guided Learning and Tutorial

Primary Actor

User

Preconditions

- The system is in a valid state of execution
- The system is not in the process of conducting a level/game

Main Success Scenario

1. User selects the tutorial option from the menu
2. System clears screen and projects empty cube
3. System chooses a cube and projects all cells
4. User clicks button to learn about octal
5. System displays popup with an octal explanation, solved examples and calculator
6. User clicks "Test" button
7. System generates random questions and textboxes for solutions
8. User enters solutions
9. System verifies answers and closes popup
10. User clicks button to learn about magic squares
11. System removes navigation controls and centres on a face (thereby a magic cube)
12. System popups explanation about magic square
13. User clicks "Test" button
14. System shows a stored magic square
15. User enters solutions

16. System verifies and shows a harder magic square
17. User exits back to main menu
18. User starts first game
19. System sets up game
20. User clicks "Hint" button
21. System popups coordinate hint about where the next cell is located
22. User clicks "Hint" button
23. System traces path to the correct square with explanations
24. User selects correct cells
25. System displays the Additon/Subtraction Box
26. User clicks "Hint" button
27. System displays the octal steps to arrive at the correct answer

Alternative Flows

- If the user asks for more magic square examples, the system displays a link to an educational website.
- If the user taps the stopwatch, the system will display a message allowing the use to disable the scoring system. If the user wants to, it will result in no scores being recorded.
- If the user runs out the level timer (22.5 minutes), the system will display a messgae saying that the score wouldn't be counted. The system will not record the score and allow unlimited mistakes and hints.

Postconditions

- The system is in a valid state of execution

5.7 UC7: Create Account

Primary Actor

User

Preconditions

- The system is in a valid state of execution
- The system is not in the process of conducting a level (can sign in between levels)
- User database should be active (in the sense that it exists and connectable) and triggerable
- The connection between the database and system should be closed.

Main Success Scenario

1. User clicks button to sign in
2. System saves current game state in browser cache and displays information to sign in
3. User selects "Create Account"
4. System displays a page with a username textbox and username guide
5. User types in appropiate username
6. System verifies username follows username requirements and pings user database
7. User database verifies username doesn't exist
8. Systems displays a page with a password textbox and password guide
9. System verifies password follows password requirements and creates a new user profile and inserts it into the user database
10. System asks user to log in
11. User logs in
12. User database confirms user
13. System recreates game state via browser cache and deletes cache
14. User completes game
15. System saves final score, time needed and date by appending it in user database
16. User database saves and closes connection

17. User goes to main menu
18. System opens connection to user database and displays information
19. User closes browser tab
20. System closes connection to user database

Alternative Flows

- If the user writes a username or password that doesn't follow the requirements, the system will display an error message alongside an option to type in another name.
- If a user database ping or save fails, the system will retry 3 times. If the system still fails to get a response to the ping, it will notify the user and say that they failed in making an account. It will also warn that any further progress might not be saved. When the user finishes the final level, the system will retry pinging the user database and if it responds, the system will display a message saying that the user creation page is working again.

Postconditions

- The connection between the user database and system should be closed
- The schema and integrity of the user database should be equivalent to its precondition

5.8 UC8: Daily Challenge

Primary Actor

User

Preconditions

- The system is in a valid state of execution
- The system is not in the process of conducting a level/game
- User and Daily Challenge database (DC) should be active (in the sense that it exists and connectable) and triggerable
- The connection between the databases and system should be closed.

Main Success Scenario

1. User clicks button to begin Daily Challenge
2. System asks if the user wants to log in
3. User logs in

4. System pings DC
5. DC replies with Daily Challenge plane colour, orientation, and starter
6. System saves Daily Challenge game state in browser
7. System asks if the user is ready to play
8. User starts challenge
9. System uses cube and starter given instead of pinging cube database and random starter
10. User finishes game
11. System pings user database and DC
12. System saves final score, time needed and date by appending it in DC column in user database
13. System creates a copyable graphic that can be displayed by the user
14. User database saves and closes connection
15. DC sends leaderboard scores to System
16. System checks if the user score belongs on leaderboard
17. System saves final score and time needed at appropiate rank by inserting it into DC's daily leaderboard column
18. System closes connection to DC
19. User goes to main menu
20. System opens connection to user database and displays information
21. User goes to Daily Challenge leaderboard
22. System pings DC for leaderboard data
23. System displays leaderboard with copy button
24. User clicks out
25. System closes DC's connection
26. User closes browser tab
27. System closes connection to user database

Alternative Flows

- If a DC ping or save fails, the system will retry 3 times. If the system still fails to get a response to the ping, it will notify the user and say that their Daily Challenge attempt will not be counted. Since the user already saw the puzzle, they will not be able to redo the Daily Challenge until it refreshes.

Postconditions

- The connection between the databases and system should be closed
- The schema and integrity of the bases should be equivalent to its precondition

6 Functional Requirements

6.1 Formal Model

This formal model describes the configuration of a game round, the evaluation of tasks, and the calculation of the users score. It defines the conditions under which a task is considered complete and the properties that the scoring system must maintain. The model supports the functional requirements concerning task execution, answer evaluation, and score calculation (FR1–FR29).

6.1.1 Game Configuration

Let the set of tasks in a game round be

$$\mathcal{T} = \{T_1, T_2, \ldots, T_8\}.$$

Each task T_i has an associated target condition and a basic score value. Let:

$$B_i = B(T_i), \qquad B_i \in \mathbb{N}_0.$$

Where B_i is the basic score awarded for correctly completing task T_i. The target conditions are defined by the game configuration and are evaluated against the current puzzle state.

The octal target values used by the game include

$$S_1 = 1001_8, \qquad S_2 = 2002_8, \qquad S_3 = 4004_8.$$

These values are interpreted as octal numbers. For example,

$$1001_8 = 513_{10}, \qquad 2002_8 = 1026_{10}, \qquad 4004_8 = 2052_{10}.$$

These values are used by the application to define their target conditions. For instance,

$$\theta_i(C) = \begin{cases} 1, & \text{if } L_i(C) = s_i, \\ 0, & \text{otherwise.} \end{cases}$$

where $L_i(C)$ is the sum of relevant cells, and

$$\theta_i(C) = \begin{cases} 1, & \text{if condition met} \\ 0, & \text{otherwise.} \end{cases}$$

is the target condition.

6.1.2 Round State

A game round is represented by the state

$$R = (C, i, A, P),$$

where:

- **Puzzle Configuration (C):** The current state of the magic cube, represented by a three-dimensional array of cell values:

  $$C \in V^{n \times n \times n},$$

  where n is the dimension of the cube and V is the set of permitted cell values, including any designated empty value used for unfilled cells. Each element $C_{x,y,z}$ represents the value of the cell at position (x, y, z), for $1 \leq x, y, z \leq n$.

- **Current Task Index (i):** The index identifying the task currently being attempted:

  $$i \in \{1, 2, \ldots, 8\}.$$

  The value of i determines which task's completion conditions are evaluated against the current configuration C.

- **Attempt Set (A):** The set of task attempts recorded during the round:

  $$A \subseteq \{(j, C_j, t_j, q_j) \mid j \in \{1, \ldots, 8\}\},$$

  where j identifies the task, C_j is the cube configuration submitted for evaluation, t_j is the elapsed time associated with the attempt, and $q_j \in \{0, 1\}$ indicates whether the attempt was unsuccessful (0) or successful (1). Each recorded attempt therefore contains sufficient information to identify the task, the evaluated configuration, its elapsed time, and its result.

- **Accumulated Score (P):** The total score awarded to the player during the round:

  $$P \in \mathbb{N}_0, \qquad P = \sum_{j=1}^{8} W_j,$$

  where W_j is the score awarded for task T_j, and $\mathbb{N}_0$ denotes the non-negative integers. A task contributes to P only when it has been completed successfully and its score has not already been awarded.

The round state must remain consistent after every accepted state transition. In particular, the accumulated score P must agree with the recorded task rewards, and unsuccessful attempts must not increase the score.

The current task index satisfies

$$1 \leq i \leq 8.$$

For each task T_i, let $E_i(C)$ be the evaluation function that returns true when the puzzle configuration C satisfies the task's required conditions and false otherwise:

$$E_i(C) = \begin{cases} 1, & \text{if task } T_i \text{ is satisfied,} \\ 0, & \text{otherwise.} \end{cases}$$

An attempt is considered successful only when the corresponding evaluation function returns 1. Changes to the puzzle state must be evaluated against the requirements of the current task.

6.1.3 Task Evaluation

Let a_i represent an attempt to complete task T_i. The result of evaluating the attempt is

$$\text{Result}(a_i) = E_i(C).$$

A successful attempt satisfies the task's target condition. An unsuccessful attempt does not satisfy that condition and must not be treated as a completed task.

The task completion indicator is defined as

$$Q_i = \begin{cases} 1, & \text{if task } T_i \text{ has been completed correctly,} \\ 0, & \text{otherwise.} \end{cases}$$

This acts as a persistent tracker of completion even after the next task begins changing $E_i(C)$.

For a round containing eight tasks, the number of completed tasks is

$$N_{\text{completed}} = \sum_{i=1}^{8} Q_i.$$

Consequently,

$$0 \leq N_{\text{completed}} \leq 8.$$

A task can contribute to the completed-task count only after its correct completion has been established by the evaluation function.

6.1.4 Timed Scoring

Let B_i be the basic score for task T_i, and let t_i be the elapsed time, in seconds, associated with its successful completion. For $t_i > 0$, the timed score is calculated as

$$\text{TimedScore}_i = \text{round}\left(\frac{B_i \cdot 1350}{t_i}\right).$$

Here, round denotes rounding to the nearest integer. The scoring rule rewards correct completion while taking the time required to complete the task into account.

The score awarded for task T_i is defined as

$$W_i = \begin{cases} \text{TimedScore}_i, & \text{if } Q_i = 1 \text{ and the score has not been awarded,} \\ 0, & \text{otherwise.} \end{cases}$$

The total score for a round is therefore

$$P = \sum_{i=1}^{8} W_i.$$

This definition ensures that the total score is the sum of the individual task scores and that an unsuccessful attempt does not contribute points. A completed task must not be awarded points more than once within the same round.

6.1.5 Formal Properties

The following properties describe expected behaviour of the task evaluation and scoring system.

Theorem 1: Correctness of Task Completion. For any task T_i, the task is considered successfully completed only if its evaluation function returns true:

$$Q_i = 1 \implies E_i(C) = 1.$$

Justification. By definition, task completion depends on satisfying the task's required conditions. An attempt that fails evaluation cannot be recorded as a successful completion.

Theorem 2: No Score for an Unsuccessful Attempt. For any task T_i,

$$Q_i = 0 \implies W_i = 0.$$

6.2 Requirements

Round setup

FR1: When the user presses START and no round is in progress, the system shall begin a round and display the 512 cells with their numbers hidden, arranged as slices of the round's plane colour.

FR2: The system shall randomly select each round's plane colour from yellow (x and y axes continuous), green (x and z axes continuous), or blue (y and z axes continuous).
Rationale: The two continuous axes form the "continuous plane" that the task definitions refer to.

FR3: The system shall randomly select one of the 384 cube orientations for each round.

FR4: The system shall require the user to choose decimal or octal as the scoring base before the starter appears.

FR5: When the user presses START after choosing the scoring base, the system shall display a randomly chosen starter number in its cell with a white background and start a timer that shows the elapsed seconds.

FR6: The system shall not reuse a starter number for a player until 192 rounds have been played, after which it shall clear that player's list of used starters.
Rationale: Keeps rounds different from each other.

Tasks and answer checking

FR7: The system shall present the eight tasks in order, starting the next task when the user presses START after completing the previous one, and shall not use the same target cells for two tasks in a round.

FR8: When the user selects a cell while a task awaits cell selections, the system shall turn the cell white if it is one of the task's target cells and red if it is not, with no limit on incorrect selections. The system shall ignore a cell that is already white and any selection made while no selection is expected.

FR9: In task 1, the system shall accept as correct the cell whose x, y, and z coordinates each differ from the starter's by 4 (modulo 8).
Rationale: This is the starter's complement. The two numbers sum to 1001 (octal).

FR10: In task 2, the system shall accept as correct the 3 cells that complete a half-orthogonal containing the starter, lying on either axis of the continuous plane and reaching an outer edge.

FR11: In task 3, the system shall accept as correct the 7 cells that complete an orthogonal containing the starter on the axis not in the continuous plane.

FR12: In task 4, the system shall accept as correct the 7 cells that complete a diagonal containing the starter and lying on both axes of the continuous plane, whether continuous or split.
Rationale: A split line needs a row, column, or pillar moved to the opposite side of the cube to appear straight.

FR13: In task 5, the system shall accept as correct the 7 cells that complete a diagonal containing the starter that uses one axis of the continuous plane and the axis not in the plane, whether continuous or split.

FR14: In task 6, the system shall accept as correct the 7 cells that complete a diagonal containing the starter that uses the other axis of the continuous plane (not the one used in task 5) and the axis not in the plane, whether continuous or split.

FR15: In task 7, the system shall accept as correct the 7 cells that complete a triagonal containing the starter, whether continuous or split.

FR16: In task 8, the system shall accept as correct the 7 cells that complete a split triagonal containing the starter that differs from the triagonal used in task 7.

FR17: Before tasks 2 to 8 begin, the system shall require the user to choose how many additional numbers to add (1 to 2 for task 2, 1 to 5 for tasks 3 to 8).

FR18: When all cells of the target line are found in tasks 2 to 8, the system shall display the starter and the chosen number of additional numbers from the line in a column for the user to add.

FR19: When the user presses DONE with an answer in the Addition Box, the system shall, if the sum is incorrect, clear the answer, turn the box red, and let the user try again, and, if correct, show the additional numbers in their cells in white.

FR20: After the target cell is found (task 1) or the addition is correct (tasks 2 to 8), the system shall ask the user to subtract from a constant: 1001 minus the starter for task 1, 2002 minus the sum for task 2, and 4004 minus the sum for tasks 3 to 8 (all octal).
Rationale: A correct subtraction shows the user found the right line, not just a lucky guess.

FR21: When the user presses DONE with an answer in the Subtraction Box, the system shall, if the answer is incorrect, clear it, turn the box red, and let the user try again, and, if correct, reset the box to white, show the remaining numbers of the line in their white cells (for task 1, show the answer in the complementary cell), and stop the timer.

FR22: If the user presses DONE with an empty answer box, or enters anything other than the octal digits 0 to 7, the system shall show a message asking for a valid answer and shall not count it as an incorrect answer.

FR23: If the user performs an input that the system does not recognize (for example, a key that has no function in the current state), the system shall show a message saying that the input was not recognized.
Rationale: Tells users why nothing happened instead of leaving them stuck (Hazard Analysis SR17).

FR24: When a new task begins, the system shall return the previous task's white cells (other than the starter) to their original colour, clear the previous question and answers, and resume the timer from its previous value.

FR25: After the subtraction in task 8 is answered correctly, the system shall end the round, display the final score, and allow a new round to begin.

FR26: Before presenting each task, the system shall check that the starter and the task's target cells add up to the correct sum (1001 octal for task 1, 2002 for task 2, and 4004 for tasks 3 to 8) and that the expected answers follow from them, and shall generate a new setup and log the failure if a check fails.
Rationale: A wrong expected answer would mark correct answers wrong and teach incorrect arithmetic (Hazard Analysis SR9, SR12).

Scoring and controls

FR27: When a task's subtraction is answered correctly, the system shall award the line points for the chosen scoring base, and shall award points for each additional number when the addition is correct, as follows (decimal / octal): task 1, 10 / 14; task 2, 20 / 30; tasks 3 and 4, 30 / 44; tasks 5 and 6, 40 / 60; tasks 7 and 8, 60 / 110; each additional number in tasks 2 to 8, 10 / 14.
Rationale: Octal points are 20% above decimal points after conversion, as in the rules.

FR28: The system shall compute each task's timed score as the task's basic score $\times$ 1350 / t, where t is the seconds spent on that task, rounded to the nearest whole number.
Rationale: The rules give the constant as 2.63671875×512, which equals 1350. Faster completion scores more. Per-task time and the rounding are our assumptions.

FR29: After each task, the system shall display the user's updated total score, and after task 8 the final score, which is the sum of the eight timed scores.

FR30: When the user presses STOP GAME during a round, the system shall end the round, erase all progress in it, record no score, and return to its ready state.

Saving and recovery

FR31: The system shall automatically save the in-progress round each time its state changes, retrying up to 3 times if a save fails and telling the user if all attempts fail.
Rationale: Prevents lost progress after a crash (Hazard Analysis SR5, SR6).

FR32: When the user returns after the page was closed, reloaded, or crashed during a round, the system shall check that the saved round is valid and offer to resume it with the timer at its saved value, discarding it and telling the user if it is invalid.

3D visualization

FR33: The system shall provide a 3D view of the cube as an alternative to the slice display, with each number in the same cell as in the slice display, and shall let the user switch between views at any time without changing the round's state or pausing the timer. Cell selection in the 3D view shall behave as in FR8.
Rationale: The 3D view is an added feature that helps users see how the cube is built (Problem Statement Goal 1). Inner cells are reached by isolating layers (FR35).

FR34: In the 3D view, the system shall let the user rotate and zoom the cube using either the pointer or the keyboard.

FR35: In the 3D view, the system shall let the user choose any single layer along any axis and show it with the other layers faded or hidden, step to the adjacent layer, and restore all layers.

FR36: The system shall highlight the chosen layer and the cells the user has selected, using visibly different highlights, and shall keep the cell states and highlighting consistent with the round's current state in both views.

FR37: If the system detects that the 3D view is running below the minimum performance level set in the Non-Functional Requirements, the system shall warn the user and offer to switch to the slice display.
Rationale: Gives users with slower devices a way to fix performance problems (Hazard Analysis SR14).

Guided learning and tutorial

FR38: The system shall provide an interactive tutorial that walks the user through a fixed sample round, requiring them to practice each of the eight task types, and shall let the user skip, exit, or replay it at any time. The tutorial shall show only instructions that match the step the user has reached, and its results shall not be recorded in statistics or leaderboards.

FR39: After an incorrect cell selection, the system shall display a message naming the type of line being sought and the rule it follows, without revealing the correct cell.

FR40: When the user requests a hint, the system shall display the coordinate step that the line in the current task follows (for example, "each coordinate differs by 4") without identifying a cell, and shall record the request.

FR41: After the user has made 3 incorrect attempts at the same step (3 incorrect cell selections in a task, or 3 incorrect answers to the same question), the system shall let the user request the solution to that step and shall record the request. For cell selection, the solution is shown by highlighting the correct cells one after another in the order the line is traced; for an addition or subtraction, it is shown as the worked octal calculation.
Rationale: Showing the answer only after repeated attempts keeps the task a challenge.

Accounts, statistics, and competitive play

FR42: The system shall let a user play without an account, and shall not send any data about a guest's play online.

FR43: The system shall let a user create an account with a unique username, a unique email address in a valid format, and a password of at least 8 characters, shall reject anything else with a message giving the reason, and shall show the entered username and email address and ask the user to confirm them before the account is created.

FR44: The system shall reject any username shorter than 3 or longer than 20 characters, or that contains a term on a prohibited-terms list.
Rationale: Usernames appear on public leaderboards, so inappropriate names could harm other users (Hazard Analysis SR13).

FR45: The system shall let a user log in and log out, and shall reject invalid credentials with a message that does not say whether the username or the password was wrong.
Rationale: Avoids revealing which usernames exist.

FR46: The system shall let a user who has forgotten their password request a reset link sent to their registered email address, without revealing whether the address is registered.
Rationale: The Hazard Analysis lists email as stored user data, and account recovery is its purpose.

FR47: When a logged-in user completes a round, the system shall record the final score, scoring base, total time, and date.

FR48: The system shall let a logged-in user view their number of completed rounds, highest score, average score, and fastest round time, for each scoring base.

FR49: The system shall display, for each scoring base, leaderboards of the 10 highest final scores and the 10 fastest round times, showing username and score or time only. A score recorded by the system shall appear on the relevant leaderboards within 1 minute.
Rationale: Separate boards by scoring base because octal scoring gives 20% more points.

FR50: The system shall create or change leaderboard and statistics records only from rounds it has completed, and shall reject any other request to do so, except for deletion under FR53.
Rationale: Prevents tampering with rankings (Hazard Analysis SR10).

FR51: The system shall offer one daily challenge each day, with the same plane colour, orientation, and starter for every user, and a leaderboard of that day's results.

FR52: After completing a daily challenge, the system shall let the user copy a summary of the date, final score, and total time that contains no cell number or answer.
Rationale: Lets users compare results without spoiling the challenge for others.

FR53: The system shall let a logged-in user delete their account, which removes their username, email, credentials, scores, statistics, and leaderboard entries.

Wellbeing

FR54: After every hour of play in a session, the system shall remind the user how long they have been playing, without changing the round's state or the timer.
Rationale: Reduces the risk of excessive use (Hazard Analysis SR21).

FR55: The system shall let the user turn off animations, including the rotation animation in the 3D view and visual effects on cell feedback, and shall apply the choice immediately.
Rationale: Reduces the risk of photosensitive seizures and eye strain from excessive or unintended flashing or animation (Hazard Analysis SR19).

7 Data Requirements

7.1 Major Data Entities

- User: a registered player.
- Cube Array: the fixed arrangement of the numbers 1 to 512 in the cube.
- Orientation: one of the 384 arrangements of the cube.
- Daily Challenge: the shared setup for one day's challenge.
- Round: one play-through of eight tasks, by a User or a guest.
- Task Attempt: the user's work on one task within a round.
- Saved Round: a saved copy of an in-progress round, used to resume it.
- Score Record: the result of a completed round.
- Used Starter List: the starters already shown to a player since the list was last cleared.
- Transaction Log: a record of the changes made to each database.

7.2 Data Dictionary

Attribute	Type	Description
User.username	Text	3 to 20 characters, unique, shown publicly
User.email	Text	Valid email format, unique, never shown publicly
User.password_hash	Text	Cryptographic hash of the password; the password itself is never stored
User.created_at	Timestamp	When the account was created
CubeArray.array_id	Integer	Identifies the array in use
CubeArray.cell_values	List of 512 Integers	Number in each cell, indexed by (x, y, z); each of 1 to 512 appears once
Orientation.orientation_id	Integer	0 to 383
DailyChallenge.date	Date	The day the challenge applies to
DailyChallenge.plane_colour	Enumeration	yellow, green, or blue
DailyChallenge.orientation_id	Integer	Orientation used
DailyChallenge.starter_value	Integer	1 to 512
Round.plane_colour	Enumeration	yellow, green, or blue
Round.orientation_id	Integer	Orientation used
Round.scoring_base	Enumeration	decimal or octal
Round.starter_value	Integer	1 to 512
Round.starter_coordinates	3 Integers	Each 0 to 7
Round.daily_date	Date	Set only if the round is a daily challenge
Round.status	Enumeration	in progress, completed, or stopped
Attempt.task_number	Integer	1 to 8
Attempt.extra_count	Integer	Additional numbers chosen: 1 to 2 for task 2, 1 to 5 for tasks 3 to 8
Attempt.selected_cells	List of coordinate triples	Cells the user selected, in order
Attempt.incorrect_count	Integer	Incorrect cell selections
Attempt.hint_requests	Integer	Hints requested (FR40)
Attempt.solution_requests	Integer	Solution requests (FR41)
Attempt.elapsed_seconds	Real	Time spent on the task
Attempt.basic_score	Integer	Points before the time formula
Attempt.timed_score	Integer	Points after the time formula
SavedRound.saved_at	Timestamp	When the round was last saved
SavedRound.state	Round state	Everything needed to resume the round, including elapsed time
SavedRound.integrity_value	Checksum	Used to check the saved round is valid (FR32)

Attribute	Type	Description
Score.final_score	Integer	Sum of the eight timed scores
Score.scoring_base	Enumeration	decimal or octal
Score.total_seconds	Real	Total time for the round
Score.achieved_at	Timestamp	When the round was completed
UsedStarter.starter_value	Integer	A starter already shown to the player
Log.timestamp	Timestamp	When the change was made
Log.operation	Enumeration	create, update, or delete
Log.record	Reference	The record that was changed

Table 5: Data Dictionary

7.3 Privacy and Data Retention

PR1: The system shall store only the data in the data dictionary, and shall not collect real names or birthdates.
Rationale: Intended users include high-school students, so collecting less personal data means less harm from a leak.

PR2: The system shall use a user's email address only for account recovery (FR46), and shall not display it or share it with any third party.

PR3: The system shall store passwords only as cryptographic hashes.
Rationale: Hazard Analysis SR2. A leak must not expose passwords that users may reuse elsewhere.

PR4: The system shall encrypt email addresses and other sensitive user data in storage.
Rationale: Hazard Analysis SR1.

PR5: The system shall check that the stored cube array contains each of the numbers 1 to 512 exactly once, and shall not start a round if it does not.
Rationale: A corrupted array would produce wrong expected answers (FR26).

PR6: The system shall automatically back up each database every 24 hours and keep the 7 most recent backups.
Rationale: Hazard Analysis SR3. The interval and count are proposed values.

PR7: The system shall log every change to each database with the time, operation, and record changed.
Rationale: Hazard Analysis SR4.

PR8: If a database is corrupted or data is lost, the system shall restore it from the most recent backup that passes an integrity check.
Rationale: Hazard Analysis SR11.

PR9: The system shall keep a user's scores and statistics until the account is deleted (FR53),

and shall then remove the user's data from live databases immediately and from backups when those backups expire.

PR10: The system shall not keep any online data from guest play (FR42), and shall delete a locally saved round when the round is completed or stopped (FR30).

8 Non-Functional Requirements

NFR1: **Cube Manipulation Performance**

The system shall update the visible cube orientation within 100 milliseconds of receiving a supported manipulation input.

Rationale: Delayed manipulation feedback can make the three-dimensional cube feel unresponsive and make it difficult for users to understand the relationship between their input and the resulting orientation.

Fit Criterion: Under the defined reference test environment, the visible cube orientation updates within 100 milliseconds for at least 95% of measured supported rotation inputs.

NFR2: **Cell-Value Editing Performance**

The system shall visibly reflect an accepted cell-value edit within 100 milliseconds of the user submitting the edit.

Rationale: Prompt feedback helps users understand whether their changes have been accepted and reduces uncertainty when solving a puzzle.

Fit Criterion: Under the defined reference test environment, at least 95% of accepted cell-value edits are visibly reflected within 100 milliseconds.

NFR3: **Puzzle Opening Usability**

A first-time user shall be able to open a puzzle without external assistance or prior training.

Rationale: Users should be able to begin solving a puzzle without needing to consult instructions or receive help from another person.

Fit Criterion: At least 80% of participants in a usability evaluation shall successfully open a puzzle without external assistance or prior training.

NFR4: **Cube Manipulation Usability**

A first-time user shall be able to manipulate the three-dimensional cube using the supported rotation controls without external assistance.

Rationale: The cube is the primary visual representation of the puzzle. Users need to understand how to change its orientation to inspect and interact with its cells.

Fit Criterion: At least 80% of participants in a usability evaluation shall successfully perform each defined cube-rotation task without external assistance.

NFR5: **Cell Selection Usability**

A first-time user shall be able to select an individual puzzle cell using the supported selection controls without external assistance.

Rationale: Users must be able to identify the cell they intend to inspect or edit. Unclear selection behaviour may cause unintended changes to the puzzle.

Fit Criterion: At least 80% of participants in a usability evaluation shall correctly select the requested cell in each defined cell-selection task without external assistance.

NFR6: **Cell-Value Editing Usability**

A first-time user shall be able to enter and submit a valid value for a selected puzzle cell without external assistance.

Rationale: Editing cell values is a core interaction in solving the puzzle. The editing process should be understandable without requiring users to learn a complex sequence of actions.

Fit Criterion: At least 80% of participants in a usability evaluation shall successfully enter and submit a valid value for a selected cell without external assistance.

NFR7: **Separation of System Components**

The system shall organize mathematical logic, user-interface logic, three-dimensional visualization logic, and online storage interactions into identifiable components with documented responsibilities.

Rationale: Separating these responsibilities reduces the risk that changes to one part of the system will unintentionally affect unrelated functionality.

Fit Criterion: A code inspection shall identify separate components responsible for each of the four specified areas, with documented responsibilities for each component.

NFR8: **Independent Testing of Mathematical Logic**

The system shall allow its core mathematical operations to be tested independently of the user interface and three-dimensional visualization.

Rationale: Independent testing makes it easier to identify mathematical errors without requiring the complete application to be executed for every test.

Fit Criterion: The automated test suite for core mathematical operations shall execute without launching the user interface or rendering the three-dimensional cube.

NFR9: **Browser Compatibility**

The system shall operate in every desktop and laptop web browser included in its defined support list.

Rationale: Users should be able to access the game through supported browsers without requiring a specific browser outside the stated compatibility requirements.

Fit Criterion: The defined core functional test suite shall pass on every browser and browser version included in the system's documented support list.

NFR10: **Browser-Based Deployment**

The system shall allow users to access and operate the game through a supported web browser without requiring installation of Python or a standalone desktop application.

Rationale: Browser-based access reduces setup effort and allows users to run the game on supported desktop and laptop devices without installing a separate application.

Fit Criterion: On a clean reference device, a tester shall be able to launch and use the game through a supported browser without installing Python or a standalone desktop application.

NFR11: **Puzzle-State Integrity**

The system shall maintain a valid in-memory puzzle state following supported cube manipulations, cell selections, and accepted cell-value edits.

Rationale: The puzzle state must remain consistent throughout interaction so that users do not encounter unintended changes or false puzzle representations.

Fit Criterion: Automated tests shall verify that the puzzle state satisfies the defined state-validity rules after every state update in the test suite.

NFR12: **Invalid Input Handling**

The system shall reject invalid cell-value inputs without terminating the application or leaving the puzzle state invalid.

Rationale: Users may enter unsupported values or make mistakes while editing cells. Invalid inputs should not corrupt the puzzle or interrupt continued use of the game.

Fit Criterion: Automated tests shall submit every defined category of invalid cell-value input and verify that the input is rejected, the puzzle state remains valid, and the application continues operating.

NFR13: **Online Operation Failure Handling**

When an online storage operation fails, the system shall inform the user that the operation was unsuccessful.

Rationale: Users need accurate feedback about whether their data has been stored or retrieved successfully. Misleading feedback may cause users to believe that their progress or puzzle data has been saved when it has not.

Fit Criterion: Tests that simulate each defined online storage failure shall verify that the system displays an appropriate failure indication.

NFR14: **Network Communication Security**

All network communication involving account credentials or user data shall use HTTPS with valid TLS.

Rationale: Encrypting network communication helps protect credentials and user data against interception or unauthorized modification during transmission.

Fit Criterion: Network inspection shall confirm that all requests involving account credentials or user data use HTTPS with valid TLS, and that the system does not transmit such information through unencrypted HTTP requests.

NFR15: **User Data Isolation**

The system shall prevent an authenticated user from accessing another user's private stored data through its supported interfaces and operations.

Rationale: Puzzle data, saved progress, and other private information associated with an account should remain accessible only to authorized users.

Fit Criterion: Authorization tests shall verify that an authenticated test account cannot retrieve, modify, or delete private data belonging to another test account through any supported operation.

NFR16: **Authentication Secret Protection**

Application-controlled persistent storage shall not store authentication secrets in plaintext.

Rationale: Storing authentication secrets in plaintext may expose user accounts if stored data is accessed without authorization.

Fit Criterion: Inspection of application-controlled persistent storage shall find no authentication secrets stored in plaintext.

NFR17: **Personal Data Minimization**

The system shall collect and retain only the personal information necessary to provide its implemented functionality.

Rationale: Limiting the collection and retention of personal information reduces privacy risks and the amount of data that must be protected.

Fit Criterion: A review of the system's data fields, collection operations, and persistent storage shall confirm that every collected or retained personal information field has a documented functional purpose.

NFR18: **Keyboard Accessibility**

All essential puzzle controls shall be operable using a keyboard without requiring pointer interaction.

Rationale: Keyboard operation supports users who cannot use a mouse or other pointer-based input device and provides an alternative way to interact with essential puzzle functionality.

Fit Criterion: A keyboard-only test shall verify that users can access and operate every essential puzzle control, including puzzle navigation, cell selection, and cell-value editing, without pointer interaction.

NFR19: **Visual Contrast**

Essential interface text and controls shall meet a minimum contrast ratio of 3:1 against their adjacent background.

Rationale: Adequate contrast helps users distinguish interface elements and read essential information.

Fit Criterion: An automated or manual contrast assessment shall confirm that every essential interface text element and control meets the specified minimum contrast ratio of 3:1 against its adjacent background.

9 AI and Machine Learning Requirements

The project does not include any AI or machine learning components, so this section does not apply.

10 Requirements Traceability Matrices

Stakeholder Req	Use Case	Functional Req
SR1	UC2	FR1, FR5, FR19, FR21, FR22, FR23, FR24, FR25
SR2	UC1, UC5	FR1, FR3, FR32, FR33, FR34, FR35, FR36
SR3	UC6	FR4, FR37, FR39, FR40, FR41
SR4	UC6	FR2, FR3, FR37, FR38, FR41
SR5	UC2	FR1, FR6, FR24, FR28, FR29

SR6	UC1	FR2, FR3, FR5, FR6
SR7	UC2, UC3	FR5, FR26, FR27, FR28

Table 6: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req
FR1	NFR1, NFR10, NFR9
FR2	NFR7
FR3	NFR7
FR4	NFR3
FR5	NFR1, NFR2, NFR9
FR6	NFR7
FR7	NFR5
FR8	NFR5
FR9	NFR6
FR10	NFR6
FR11	NFR12
FR12	NFR2
FR13	NFR18
FR14	NFR11
FR15	NFR19
FR16	NFR19
FR17	NFR11
FR18	NFR2
FR19	NFR2, NFR12, NFR9
FR20	NFR8
FR21	NFR11, NFR12
FR22	NFR12
FR23	NFR11
FR24	NFR7
FR25	NFR8
FR26	NFR8
FR27	NFR8
FR28	NFR2

FR29	NFR11, NFR17
FR30	NFR13
FR31	NFR13
FR32	NFR18, NFR9
FR33	NFR4, NFR9
FR34	NFR1, NFR4, NFR9
FR35	NFR19, NFR9
FR36	NFR18, NFR19, NFR9
FR37	NFR3
FR38	NFR2
FR39	NFR2
FR40	NFR2
FR41	NFR2
FR42	NFR14, NFR17
FR43	NFR14, NFR16
FR44	NFR12
FR45	NFR14, NFR16
FR46	NFR14, NFR16
FR47	NFR15
FR48	NFR15
FR49	NFR15
FR50	NFR11, NFR15
FR51	NFR17
FR52	NFR17
FR53	NFR15, NFR17

Table 7: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

11.1 Scope

- **Order 8 cube (Section1.2):** An order 8 cube is chosen as it is the smallest most perfect magic cube.
- **Fixed cube and no player generation (Section1.2):** The cube being used is fixed, with players being unable to generate their own, as a single cube can generate a large number of puzzles and would massively balloon the scope and difficulty in assuring all cubes are perfect magic cubes.
- **Browser based play (Section1.2):** The game is strictly web-based as we believe a

browser-based game is sufficient to reach a large enough number of our intended users; browser play maximizes access across devices and locations.

11.2 Functional Requirements

- **Round setup** (FR2, FR3, FR6):
  The plane colour, cube orientation, and starter are chosen at random, and a starter is not reused for 192 rounds, as set out in Dr. Rapoport's rules. This keeps rounds different from each other so players cannot memorize one arrangement, which supports the goal of a different set of tasks each round.

- **Tasks and answer checking** (FR9–FR16, FR19, FR21):
  The eight tasks, their target cells, and their order from easiest to hardest come directly from Dr. Rapoport's rules rather than from our own design. The addition and subtraction steps show that a user found the right line instead of clicking cells by chance, and they give practice with octal arithmetic.

- **Task verification and input handling** (FR7, FR26, FR22, FR23):
  Each task is checked before it is shown, because a wrong expected answer would mark correct answers as wrong and teach incorrect octal arithmetic (Hazard Analysis SR8, SR9, SR12). Invalid or unrecognized input produces a message instead of counting as a mistake, so users are not penalized for typing errors or left stuck (SR17).

- **Scoring** (FR27, FR28, FR30):
  The point values, the 20% higher points for octal, and the time formula come from Dr. Rapoport's rules. We state the time constant as 1350, which equals the rules' 2.63671875×512, so that it is easy to check and test. A stopped round records no score, so abandoning a round cannot change a leaderboard.

- **Saving and recovery** (FR31, FR32):
  Progress is saved after every change and a failed save is retried three times, so a crash or lost connection does not cost the user a round (Hazard Analysis SR5, SR6). A saved round is checked before it is loaded, so a corrupted save cannot crash the game or load an incorrect state (SR7).

- **3D visualization** (FR33, FR35, FR36, FR37):
  The 3D view is a feature we added to the game (Problem Statement Goal 1). The slice display from Dr. Rapoport's rules stays available, and each number stays in the same cell in both views. Layers can be isolated so users can reach inner cells that other cells hide. Both views must show the same cell states (Hazard Analysis SR15), and users on slower devices are warned and offered the slice display (SR14).

- **Guided learning and tutorial** (FR38, FR39, FR40, FR41):
  These requirements come from the Problem Statement goal of teaching octal arithmetic and the cube's patterns while users play. A hint gives the coordinate step without

naming a cell, and the full solution is available only after 3 incorrect attempts at the same step, following feedback from Dr. Rapoport, so help does not remove the challenge of the task. Tutorial steps and hints must match the user's current step (Hazard Analysis SR20).

- **Accounts and competitive play** (FR42, FR43, FR44, FR49, FR50, FR53):
  Online play comes from the competitive goal in our Problem Statement. Guest play sends no data online, so students can try the game without an account. Usernames appear on public leaderboards, so they are restricted (Hazard Analysis SR13). Leaderboards are separate for each scoring base because octal scores are 20% higher, and only the system can create or change leaderboard records so rankings stay fair (SR10). Because we store usernames and email addresses, users can delete their account and data.

- **Daily challenge** (FR51, FR52):
  Every user receives the same setup each day so results can be compared, which gives users a reason to return. The shared summary leaves out cell numbers and answers so that sharing a result does not spoil the challenge for others.

- **Wellbeing** (FR54, FR55):
  Both come from the Hazard Analysis. The hourly reminder reduces the risk of excessive use (SR21), and the option to turn off animations reduces the risk of photosensitive seizures and eye strain (SR19).

11.3 Data Requirements

- **Game data** (cube array, orientation, round, and task attempt):
  The cube array holds each of the numbers 1 to 512 exactly once, as in Dr. Rapoport's documents, and has an identifier because those documents say the array may change. It is checked when loaded (PR5), because a corrupted array would produce wrong expected answers. Rounds and task attempts are stored separately so each task's time, mistakes, hints, and score can be recorded, which the scoring, statistics, and tutorial requirements depend on.

- **Saved rounds**:
  A saved round includes an integrity value so it can be validated before it is loaded (FR32).

- **Statistics and leaderboards**:
  These are calculated from score records instead of being stored separately, so the same information is not kept in two places and cannot disagree.

- **Personal data** (PR1–PR4):
  Only a username, an email address, and a password hash are stored, with no real names or birthdates, because intended users include high-school students. The email

address is used only for account recovery. Hashed passwords and encrypted data limit the harm if the database is leaked (Hazard Analysis SR1, SR2).

- **Backups, logging, and retention** (PR6–PR10):
  Backups, change logs, and restoring from the last valid backup protect against lost or corrupted data (Hazard Analysis SR3, SR4, SR11), although the Hazard Analysis defers them until after the main features. Data is kept until the user deletes their account, and guest play is never stored online, since there is no reason to keep it.

11.4 Nonfunctional Requirements

- **Performance (NFR1, NFR2):**

  Cube manipulation and cell-value editing are two of the most frequent interactions in the magic cube game. These operations need to provide prompt feedback so that users can understand the effects of their actions without unnecessary delays.

- **Usability (NFR3, NFR4, NFR5, NFR6):**

  The usability requirements focus on the primary tasks that users must perform to interact with the puzzle successfully. These tasks represent the essential interaction workflow. Requiring at least 80% of usability evaluation participants to complete these tasks without external assistance provides a measurable baseline for whether the interface is understandable to its intended users or not.

- **Maintainability (NFR7, NFR8):**

  Separating mathematical operations, user-interface behaviour, 3D visualization, and online storage into identifiable components reduces coupling between unrelated responsibilities. This decision allows individual components to be modified with less risk of affecting other parts of the system.

- **Portability (NFR9, NFR10):**

  The system is intended to be accessed through desktop and laptop web browsers rather than a platform-specific desktop application. Browser-based deployment reduces installation requirements and allows users to access the game through supported environments without installing Python or additional standalone software.

- **Reliability (NFR11, NFR12, NFR13):**

  The correctness of the puzzle state is essential because subsequent calculations, visual updates, and user interactions depend on it. Separately defining requirements for valid state maintenance, invalid input handling, and online failure handling ensures that the system can respond predictably to both expected operations and error conditions.

- **Security and Privacy (NFR14, NFR15, NFR16, NFR17):**

Because the system may support user accounts and online storage, appropriate measures are needed to protect user information. Encrypted network communication protects data during transmission, while access restrictions prevent users from accessing other accounts' private data.

- **Accessibility (NFR18, NFR19):**

  The system should provide alternative ways for users to interact with and understand the puzzle. Keyboard-operable controls support users who cannot rely on pointer-based interaction, while sufficient visual contrast helps make interface elements distinguishable.

12 Likely Changes

(NFR1, NFR2) : If it is found that the 100ms response times are unreasonable can easily be changed to more fitting ones.

(NFR19) : May need to change the given standard for contrast if it is not considered enough.

LC1: Additional number bases and their corresponding order cubes may be addressed after the project meets all current specifications, changing the scope laid out in Section1.2.

LC2: Section1.2's scope may also change in the form of the addition of new game modes and rulesets such as partially filled cubes, offline modes, and player puzzle generation.

(UC3) : The team still aren't really aware of how they want the scoring mechanism to work as timing doesn't feel like the funnest scoring method. So far, they followed Dr. Rapoport's idea, but this might change after testing from stakeholders. Also, a lot of the work the system does can be offloaded to the scoring system, as the scoring system is just a glorified time signal in this use case.

(UC4) : There might be too much saving involved in this, as the internal object representing the game in the system is saved before being changed on the front end (which the user sees) every single time. It might be useful to use the browser cache so data is saved only at major steps.

(UC5) : The team still hasn't started learning about 3D visualization and how it translates to a browser, so extensive QA is needed to arrive at an approachable cube visualization.

(UC8) : There are multiple ideas on how the Daily Challenge is going to be implemented, and since it is a strech goal and the team doesn't really know what aspects of their project is "fun", it can be a vastly different implementation on release.

13 Unlikely Changes

(NFR18) : Will not likely be lessening our goal to allow those without the ability to use a mouse to be able to interact with the program.

(NFR9, NFR10) : The web based approach was one of the first comments made by Dr. Rapoport and so unless there is no other option we would like to continue with this method.

ULC1: Section1.2's scope is unlikely to change to support mobile app releases as the current medium was agreed upon to be sufficient by both the team and the sponsor.

ULC2: Section1.2's scope is unlikely to change to include features of monetization as that defeats the purpose of the game's creation.

(UC1, UC2, UC7) : These describe core aspects of the magic 8 project, so they are very unlikely to change as there were numerous meetings to arrive at these. Obviously, the rules of how magic cubes and octal is not going to change, so the use cases here are locked for now. Account creation is unlikely to change as numerous components depend on it, so changing its workflow would require significant changes throughout the codebase.

(UC6) : Whilst this might not be optimal way to teach the subject matters related to our project, it is the best the team currently came up with. Also, as it is a stretch goal, there isn't going to be a lot of focus on it until the principle project is finished.

14 Appendices

14.1 Appendix A: Additional Materials

N/A

14.2 Appendix B: Development Plan

This section describes the phased implementation for the functional requirements listed in the document; it is organized around hedging the major risks of 3D visualization and the base game's core gameplay sequence.

14.2.1 Phase 1: Proof of concept

Phase 1 is the POC, a hardcoded, minimal version of the game aimed at validating the two main risks. The requirements involved in this part are:

- Round setup: FR1, FR2, FR3, FR4, FR5
- cell interaction: FR7, FR8, FR9, FR10, FR11, FR12, FR13, FR14, FR15, FR16, FR17, F18
- 3D Visualization: FR32, FR33, FR34, FR35. FR36 is excluded as it is an extension of FR36 aimed at improving accessibility.
- Task implementation: FR19, FR20, FR21, FR22; these are all checks that should be available even in a hardcoded version of tasks.

Success criteria: The cube fully renders and can be navigated according to the 3D visualization requirements; the user must be able to complete all 8 tasks without logic issues. If the 3D visualization fails, a transition to a different implementation method or 2D must be evaluated.

14.2.2 Phase 2: Core game loop implementation

This phase is aimed around randomization of the round setup and completing the scoring and task evaluation system. The requirements involved in this part are:

- Round setup: FR1, FR2, FR3, FR4, FR5 must still hold and FR6 is added to ensure randomization constantly changes the starting voxel.
- Task Execution: FR19, FR20, FR21, FR22 must all still hold and FR23 and FR24 will be used to verify the task works with a randomized starting voxel.
- Integrity check: FR25 to verify expected answers and verify the validity of task lines.

- Scoring: FR26, FR27, FR28 which will revolve around verifying the points scoring, time-based scoring, and score display.
- Stop and reset: FR29 to ensure the game loop can be stopped.

Success criteria: A full round of 8 randomized tasks must be playable from start to finish. The system must correctly accept valid answers and update the user's score while rejecting invalid or incorrect ones. The integrity checks must pass on every generated task, scores must be computed and displayed on every round, and the STOP GAME button must end the round and erase that round's progress.

14.2.3 Phase 3: Improved 3D visualization and Navigation

This phase is aimed at upgrading the 3D visualization implemented in the game, improving usability and accessibility. The requirements involved in this part are:

- View Switching: FR23
- Navigation: FR33
- Layer control: FR34
- Highlighting: FR35
- Accessibility: FR36

Success criteria: Users must be able to switch between slice and 3D views at any time without progressing the round state or pausing the timer; rotation and zoom must work with both pointer and keyboard input. Any layer along the axis must be selectable, interactable, and restorable, with chosen layers being highlighted. The four defined appearances must be distinguishable via means other than just color, and at least 80% of users in the usability test must be able to perform rotation and layer selection without assistance.

14.2.4 Phase 4: Learning phase

This phase focuses on adding the educational aspect and guided learning through the tutorial and explanation features. The requirements involved in this part are:

- Interactive tutorial: FR37
- Mistake explanation: FR38
- Worked calculations: FR39
- Hints: FR40 and FR41

Success criteria: A first-time user must be able to complete the interactive tutorial on their own and practice all eight tasks. The skip, exit, and replay features must be functioning. The hint system should display a message naming the task/line type and its rules without revealing the correct cell, and the worked calculations should be available on request after answering incorrectly.

14.2.5 Phase 5: Online feature phase

The goal of this phase is to implement all the features with the creation of user account creation/deletion, leaderboards, round records, and daily challenges. The requirements involved in this part are:

- Guest Play: FR42.
- Account Management: FR43, FR44, FR45, FR46.
- Statistics: FR47, FR48.
- Leaderboards: FR49, FR50.
- Daily challenge: FR51, FR52.
- Account deletion: FR53.

Success criteria: Users must be able to play without an account with none of their data being stored. Account creation must enforce unique usernames and emails that meet the prohibited term rules, a password that meets the safety definition; invalid inputs must be rejected with a reason, with the reset feature working without revealing which credential was wrong. Completed rounds must record the user's final score, scoring base, total time, and date; statistics must be displayed and leaderboards must show at least the top 10 scores and times with the username. Leaderboards and statistic records must be created only from complete rounds and reject any other requests; the daily challenge must present the same challenge to every user with an accompanying summary. Account deletion must remove all user data, verified by post-deletion queries.

14.2.6 Phase 6: Recovery phase

This phase is focused on increasing the system's resilience and interruption handling capability. The requirements involved in this part are:

- Auto-save: FR30
- Round restoration: FR31

Success criteria: Any round in progress must be saved automatically on each state change, with the system trying at least 3 times upon save failure and notifying the user if all attempts fail. After a page closes, reloads, or crashes, the system must check the saved round's validity and offer the user the option of resuming the timer at its saved value while discarding improper saves and informing the user. Automated tests should simulate the state save failures and invalid saved states to confirm retry behavior, user notification, and safe discard.

This phase is deferrable if timeline limitations apply as the game is fully playable without it.

14.2.7 Phase 7: Stretch goal implementations

After the completion of the initial system, the team will be free to explore out-of-scope stretch goals such as the following:

- Creation of cubes in other number bases.
- Alternative rulesets and objectives such as partial square completion rounds.
- Automatic generation of new magic cube formations.
- Player puzzle generation.
- Offline play.

14.3 Appendix C: Values of Auxiliary Constants

N/A

14.4 Appendix D: Generative AI Disclosure

A model used was DeepSeek-V4.1-Flash

Generative AI was used to brainstorm a list of functional requirements and data requirements for our project (Sections 6 and 7). The rules and player instruction documents provided by Dr. Rapoport were used as context for the AI model to ensure the list of requirements was comprehensive and accurate to the game rules. Generative AI was also used to update the requirements to reflect feedback from Dr. Rapoport. In addition, it was used to draft the rationale for the functional requirements and data requirements in section 11. The AI model that was used was Claude Sonnet 5.5, with Medium effort. All AI-assisted content related to requirements and rationale was checked against Dr. Rapoport's documents and our Hazard Analysis before being included. For sections 2, 8, and 6.1 there was AI used for text generation using a particular format from unformatted text due to my unfamiliarity with LaTeX. For section 1 AI was used to suggest user inputs and outputs an provide insight into what laws and regulations the game would have to comply with and to create the tables for the mentioned sections. for Section 10 AI was used to create the table structures.

14.5 Appendix E: Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

Team Answers

Questions 5 and 6 have been answered as a team.

5. There are a few key pieces of knowledge and skills that the team will collectively need to acquire to successfully complete this capstone project. First, we must further our

understanding of magic cubes and the patterns that are present within them so that we can accurately create the game which Dr. Rapoport has asked us to create. Second, we must learn how to do 3D projection onto a 2D plane to allow for us to display a cube using only pygame which is required for our current plan to use a python to WebAssembly translation method. Third, we will need to learn data storage methods and how to effectively send and retrieve data that may need to be encrypted for privacy. In order to make sure the rules are formalized properly, the team might have to use functional programming languages (like Lean) to formally prove that their rules are stringent.

6. Josh will be taking on the task of learning 3D projection on a 2D plane. The two ways in which he will go about doing this are using a template version of the code for a simple object and studying it to gain further understanding of the concepts as well as watching instructional videos on the topic from online sources such as youtube. He mainly plans to focus on breaking down the code since that will be sure to work with pygame while some videos that are found can be witholding crucial information.

   Jacob will be completing the task of furthering the team's understanding of magic cubes and their patterns. The first method Jacob will use is discussions with the supervisor, Dr. Rapoport. Dr. Rapoport is already very knowledgeable on magic cubes and their properties as well as the context of the project. As such, Dr. Rapoport will be a very useful source of information. The second method Jacob will use is literature review. Magic cubes and their properties are well documented in academic literature as well as other online sources (e.g., websites). Jacob will use this literature review to supplement knowledge gained from discussions with Dr. Rapoport.

   Allen will be taking on the task of learning about data storage methods, specifically databases. The first method Allen will use is researching and comparing different database options, such as their features, cost, and how well they work with the team's web-based plan, to determine which is the best fit for the team. The second method Allen will use is reading documentation and following tutorials for the chosen database to learn how to store and retrieve user accounts, scores, and leaderboard data.

   Nehan will be learning how to formally prove and keep track of all possible instances of magic cubes. This might require building a seperate plugin that is useful for proving magic cube-related or octal-related logic. If this method fails (shouldn't due to Nehan taking Functional Programming as an elective), he will find a way to generate all valid possible states of the cube and having the system look up to see if the user-entered cube exists.

Josh Engel

1. One thing that went well with writing this deliverable is compared to the last deliverable we were more organized on what we wished to complete and when, as well as starting sooner allowing for easier communication throughout the group. I also used

the provided materials from earlier years as a guide more than with the last deliverable which helped to identify what exactly I should be doing.

2. The main pain point I found throughout this deliverable was how to make NFRs verifiable. I was unsure as to what would be considered acceptable for fit critereon and so changed formatting and explanations a few seperate times leading to uneeded time spent. Eventualy I came up with a very simple solution which was to reach out and "poll" my aquaintances as well as teammates on what they deemed would be acceptable and proceeded from there.

3. Personally I worked on the Non-functional Requirements as well as the formal model. For the NFRs I based them off earlier conversations we had with our supervisor Dr. Rapoport as well as getting feedback from him for changes after the initial draft. For the formal model I used the functional requirements determined by my peers as well as had a peer review of it to help reach a state the group was happy with.

4. Multiple courses I have taken will be of great help to us throughout this project. A major one of these courses is 4HC3: Human and Computer Interfaces, this will help in designing the applications UI in a human forward fashion which allows for better usability this is especially important for me since I am currently slated to be working on front end development where this is important.

Amos Edeaghe

1. While writing this deliverable, we were able to get a lot of feedback from our supervisor as we started work on it early, making the process feel collaborative. The early start was very useful in giving us time to discuss our sections with each other and seek guidance on how to define aspects such as scope.

2. The most challenging point for me was defining the scope of the project in a way that fully encapsulated the idea we had for the game. I tackled this through the feedback sessions with our supervisor, my teammate Jacob, and the notes I had from previous meetings with the team. I also found the dependency between sections difficult as our initial split did not fully account for this, but communication with my teammates on what sections I needed prevented this from growing into a major issue.

3. As I worked on the scope first section, I can say essentially the entire scope, its stakeholders, and its expected inputs were created by discussing with our supervisor and with our peers, with some of its aspects being refined with input from previous capstone students who have worked with Dr. Rapoport.

4. A lot of my courses this year are capstones such as ENGMGT 5B03 and COMMERCE 4PA3 courses which all focus heavily around the non-implementation aspects of the project, meaning I always get to experience the specification process from multiple angles when working on this course. I've also taken courses such as Human Computer

Interfaces and Software Design that focus around ensuring a final implemented product meets specifications and appeals to users; I think all of these will be useful in working on this project.

Nehan Munir

1. One thing that went well with writing this deliverable is that the development plan helped us focus in on an idea that we can make quantifieable steps for. Speaking with Dr. Rapoport over Reading Week helped ensure us that we were following Dr. Rapoport's vision (as he is a stakeholder for this project). This made it easier to split up work as we all knew that everyone was on the same page from the offset, while we needed numerous meetings beforehand to land on a stable idea for the Capstone project.

2. Whilst the previous section might make it seem like there wasn't really any conflict on the idea, we still needed a level of connectivity between every section in the project. For example, we needed to be clear how stakeholder requirements lead to the FRs and NFRs connect, as well as the connection between use cases and the functional requirements. Getting the appropiate hyperlinks was difficult without knowing exactly what the other person wrote and for which point. This was solved by essentially finishing our work before meeting with Dr. Rapoport, allowing us ample time to reread everyone's work and see the connectivity of ideas.

3. As I worked on the stakeholder requirements, I came up with 8 solid ones based on a file that Dr. Rapoport wrote explaining how magic squares work and how he wants the project to work. I was then able to ask him to look them over during our Reading Week meeting. There, we realized that we didn't explain our strech goals to him (ex. tutorials on octal and magic squares), and we were able to get them approved, thus allowing us to add it as stakeholder requirements.

4. I will be working on formalizing how magic squares will work (the rules) within the system. I feel like SFWRENG 2DM3/2FA3 - Discrete Mathematics with Applications I-II and Linear Algebra I will be important. Discrete Math allows me to ensure that the cube is always in a steady state and map out all of the possibilites the cube can take. Linear Algebra will help as we are likely going to represent all 512 values inside the cube as a matrix, and being able to create formal rules on how the matrix can be manipulated is important (whether it is a database check to ensure all the cube matrixes are actually magic cubes, or an instaneous check on what the user is currently entering).

Allen Tong

1. What went well for me was having Dr. Rapoport's rules and player instruction documents to work from. The eight tasks, scoring system, and answer-checking steps were

already specified, so I was able to write most of the base game requirements directly from them. In addition, having Dr. Rapoport review sections 6 and 7 went well, since his comments caught problems I would not have noticed.

2. My main pain point was understanding the game well enough to write precise requirements. I resolved this by writing the requirements one task at a time, checking each against the rules and instruction documents, and having sections 6 and 7 reviewed by the team and Dr. Rapoport to catch any mistakes in my understanding of the rules.

3. About half of the functional requirements in section 6 came from Dr. Rapoport, who is both a stakeholder and our supervisor. These requirements were based on his rules documents and the meetings we had with him as a full group.

4. SFWRENG 3RA3 will help the most because it taught us how to write precise and verifiable requirements. This is important because the SRS is the basis for everything we build and test. Requirements that are clearly written are easier to turn into test cases, and they prevent misunderstandings with the client in the future. SFWRENG 4HC3 will also help because it covers evaluating interactions and designing for users. This applies directly to our tutorial and hints components, which have to be usable for our intended users.

Jacob Brodersen

1. I did not write any of the sections for this deliverable (I focused on the Hazard Analysis). However, the crossover of the two documents (referencing the requirements from the Hazard Analysis) went well because the group kept in communication to ensure the two documents were consistent.

2. I did not write any of the sections for this deliverable (I focused on the Hazard Analysis). However, one pain point was trying to figure out how to incorporate LO_SpecMath into this document. We had difficulty deciding the format and location of the content. To resolve this, Josh (the writer of the formalization section) and I discussed various ideas. We ended up deciding upon a method suggested in the Discussion page on Avenue because Dr. Smith commented that he liked the team's approach.

3. Most of our functional requirements came from our supervisor, Dr. Rapoport. In terms of the other types of requirements (i.e., non-functional requirements, safety and security requirements, etc.), these were brainstormed internally.

4. I think the SFWRENG 3RA3 has been helpful already in this course. Stating the obvious, it was helpful for writing the requirements in the Hazard Analysis and SRS documents. Furthermore, the 3RA3 course knowledge was helpful in the Problem Statement document for defining the problem.

I think that SFWRENG 4HC3 will become helpful when we start thinking about interface design; in this course, we did a full interface design for our own app idea. This is very similar for what we will have to do in this capstone.

In SFWRENG 3A04 we were able to design a software architecture and develop the code for a classification application. This will help us in the design and development phases of this capstone.

Finally, SFWRENG 2AA4 gave us practical experience in software design and development, which will help in this capstone when creating the application.